



Egyptian National Consensus on Dermocosmetic Ingredient Selection Across Common Dermatology Scenarios: A RAND/UCLA Appropriateness Study

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ABSTRACT

Introduction: Dermocosmetics are widely used to complement dermatologic care, yet context-specific guidance remains limited for populations with Fitzpatrick skin types III–V.

Prior Presentation: Preliminary findings from this consensus were presented at the International Cosmetic Congress & Exhibition (ICCE), Cairo, Egypt, 17–19 September 2025. The present work contains the full dataset and complete analyses. The manuscript has not been previously published in a peer-reviewed journal.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s13555-025-01650-5>.

We convened a national expert panel to generate transparent, reproducible recommendations across ten common clinical scenarios.

Methods: Egyptian dermatologists participated in round 1 (national survey, $n=601$) and round 2 (expert panel, $n=16$), both with anonymous ratings and inter-round feedback, using the RAND/UCLA appropriateness method (median bands 1–3/4–6/7–9; disagreement index (DI)=interpercentile range (IPR)/IPR adjusted for symmetry (IPRAS); $DI > 1.0$ =disagreement). Per-item outputs included median, P30, P70, IPR (30–70), asymmetry Index (AI), IPAS, DI, and final category (appropriate/uncertain/inappropriate). We additionally benchmarked classifications against

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prior consensus, guidelines, and key evidence frameworks.

Results: Across ten vignettes and 30 ingredients (30 ingredients $\times$ 10 scenarios = 300 items; multiple raters per item in round 1), 158 (52.7%) items were appropriate (median ≥ 7 ; $DI \leq 1.0$), 135 (45.0%) were uncertain, and 7 (2.3%) were inappropriate. Photoprotection had the highest appropriateness across scenarios (broad-spectrum/tinted SPF), with a small number of DI -flagged uncertain exceptions. Hydration/barrier agents (e.g., hyaluronic acid, peptides, ceramides) were appropriate in stress-aging, post-laser, and post-procedure care. Pigment modulators (tranexamic acid, arbutin, niacinamide, vitamin C, glabridin) were appropriate in melasma/post-inflammatory hyperpigmentation and chronic sun-induced pigmentation. Classical retinoids were inappropriate for postpartum/breastfeeding and immediate post-procedure; lower-irritancy retinoid esters were context-dependent. Botanicals showed inconsistent support. Panel disagreement ($DI > 1.0$) declined from 38.3% in round 1 to 15.7% (47/300) in round 2. Patterns largely aligned with prior consensus; visible-light-mitigating photoprotection and timing-specific retinoid use were emphasized for darker phototypes.

Conclusion: We provide a transparent, regionally relevant framework for dermocosmetic ingredient selection. Sun protection and barrier support are foundational; pigment modulators are scenario-specific; retinoids require selective use; botanicals remain adjunctive.

PLAIN LANGUAGE SUMMARY

Doctors and patients often use “dermocosmetics” (medical-grade skincare) to support treatments for dark spots or patches (melasma, post-acne marks), aging, or recovery after procedures. But it has not been clear which ingredients fit which situations—especially for people with darker skin types living in sunny climates. We brought together dermatology experts from across Egypt to rate 30 ingredients (for example sunscreen, hydrating agents, vitamin C, tranexamic acid, and retinoids) across 10 everyday scenarios such as melasma after pregnancy, post-inflammatory hyperpigmentation after acne, post-laser care, and stress-related skin aging. We used a structured method that keeps individual ratings anonymous and sets rules in advance for what counts as “appropriate,” “uncertain,” or “inappropriate.” We also measured when the experts disagreed and reported those details to keep the process transparent. In summary, sunscreen was essential across scenarios—especially tinted options that also block visible light. Hydrators and barrier-repair ingredients (like hyaluronic acid and ceramides) were a strong base for stressed or recently treated skin. Brightening ingredients (like tranexamic acid, arbutin, niacinamide, and vitamin C) helped most when pigmentation was the main concern. Retinoids were helpful for aging in suitable patients but not right after procedures or while breastfeeding. Botanical extracts had mixed support and are best used as add-ons. This guidance

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can help tailor skincare choices to real-world patients and skin tones.

Keywords: RAND/UCLA; Appropriateness; Dermocosmetics; Consensus; Skin of color; Egypt; Hyperpigmentation

Key Summary Points
<i>Why carry out this study?</i>
Dermocosmetics are widely used for pigmentation, aging, and post-procedure recovery in darker skin types, but context-specific guidance is limited.
We used the RAND/UCLA framework to develop transparent, regionally relevant guidance for ingredient selection across common Egyptian dermatology scenarios.
We situate the recommendations against prior consensus to highlight what is foundational vs. scenario-dependent for darker phototypes in high-UV settings.
<i>What was learned from the study?</i>
Sunscreen showed high appropriateness across scenarios, with a few DI-flagged uncertain exceptions; hydration/barrier-repair ingredients were appropriate in stress-aging and post-procedure/laser care; brighteners (tranexamic acid, arbutin, niacinamide, vitamin C, glabridin) were appropriate when pigmentation dominated.
Classical retinoids were not appropriate immediately after procedures or during breastfeeding; botanicals showed inconsistent support; these rules enable tailored dermocosmetic choices and motivate prospective validation.

INTRODUCTION

Dermocosmetics—topical products that combine cosmetic and biologically active ingredients

designed to act beyond simple “cosmetic” effect—have become increasingly relevant in dermatological practice. They aim not only to improve aesthetic appearance but also to support skin barrier function, modulate inflammation, and complement medical therapies (e.g., in acne, pigmentation, and post-procedural settings) [1, 2].

In antiaging dermocosmetics, an international consensus recently emphasized that dermocosmetics may be used both as monotherapy in milder forms and as adjunctive therapy to enhance tolerability, reduce lesion counts, and maintain treatment gains [3].

One key area of demand is hyperpigmentation and pigmentary disorders, which are particularly frequent among darker-skinned populations and have significant psychosocial impact. Dermocosmetic strategies for pigmentation often target multiple steps of melanogenesis—such as tyrosinase inhibition, antioxidant activity, melanosome transfer modulation, and enhancement of epidermal turnover—while maintaining a favorable safety profile. Recent advances have also emphasized the importance of melanosome transport and keratinocyte-mediated pigment processing as critical control points in pigmentation disorders [4–6].

Despite the theoretical and practical rationale for dermocosmetics, gaps persist in regionally validated, consensus-based guidance—especially in populations with higher Fitzpatrick skin types or unique environmental exposures. Many published recommendations derive from Western or East Asian populations; fewer are calibrated to Middle Eastern or North African skin biology, phototypes, and patient preferences [4, 5, 7]. Moreover, while consensus methodology has been applied in the context of antiaging dermocosmetics [3], comprehensive consensus evaluating a broad panel of dermocosmetic ingredients across diverse clinical scenarios remains rare.

To address this gap, we conducted a nationwide, RAND/UCLA appropriateness study among Egyptian dermatology experts, using ten representative clinical vignettes drawn from real-world practice. Our objectives were to (1) gauge consensus on the suitability of various dermocosmetic active ingredients in distinct clinical contexts, (2) characterize patterns of

agreement or divergence across cases, and (3) propose a context-sensitive dermocosmetic recommendation framework applicable to dermatologists in populations with higher Fitzpatrick skin phototypes.

In doing so, this work seeks to contribute a rigorously derived, regionally relevant guideline to support evidence-informed dermocosmetic decisions in daily dermatology practice.

METHODS

Study Design and Framework

We conducted a structured national consensus using the RAND/UCLA appropriateness method (RAM) to evaluate dermocosmetic ingredient selection across common dermatology scenarios. RAM combines independent expert ratings, controlled feedback between rounds, and a priori classification/disagreement rules to yield transparent, reproducible judgments when randomized evidence is limited [8]. This study was conducted in accordance with the Declaration of Helsinki and approved by the Ethical Committee of the Faculty of Medicine, Misr University for Science and Technology.

Clinical Case Scenarios

Ten standardized clinical vignettes were designed to reflect routine dermatology consultations in Egypt (post-pregnancy skin recovery; stress-induced aging; post-procedure skin-care; post-inflammatory hyperpigmentation [PIH]; rosacea-prone skin; lentigines and UV rays; post-acne rejuvenation; celebrity concerns [time-sensitive public-facing]; smoker with premature aging; bridal skin rejuvenation [pre-event preparation]). Two vignettes were framed as time-sensitive, high-visibility consultations (media/public-facing and pre-event preparation), which commonly arise in practice and may influence tolerability considerations and timing of active ingredients. Each vignette specified patient demographics (age, sex, Fitzpatrick phototype), relevant history, and key clinical findings. Vignettes were developed by a core

scientific committee and based on recurring real-world consultations and common problem-goal combinations (e.g., pigmentation, sensitivity/rosacea, post-procedure care, acne/PIH, photoaging). Draft vignettes were iteratively refined for clarity and clinical realism and pretested prior to fielding; no rating results were discussed during pretesting.

Ingredient Set

Dermatologists rated 30 actives, organized into five functional domains, in every scenario (unit of analysis = $30 \times 10 = 300$ items per round):

- Photoprotection ($n=2$): broad-spectrum sunscreen; tinted sunscreen (iron oxides)
- Brightening/antioxidants ($n=11$): niacinamide; arbutin; tranexamic acid; kojic acid; azelaic acid; glutathione; glabridin; Lumiskin (diacetyl boldine; trade name illustrative); vitamin C; vitamin E; vitamin K
- Regenerative/hydration/barrier ($n=7$): hyaluronic acid; PrimalHyal ultra-filler spheres (cross-linked hyaluronic acid micro/nano-spheres; trade name illustrative); collagen; elastin; peptides; ceramides; squalane
- Renewal/retinoids/exfoliants ($n=4$): glycolic acid; salicylic acid; classical retinoids; granactive retinoid (hydroxypinacolone retinoate)
- Botanicals/other ($n=6$): aloe vera; *Centella asiatica*; arnica; snail extract; tea tree oil; caffeine

Panel Selection and Composition

Purposive sampling recruited dermatologists across academic and community/private settings from multiple regions of Egypt with recognized expertise in cosmetic/laser dermatology. Participants were Egyptian nationals in independent practice for ≥ 5 years, with $\geq 30\%$ of their clinical workload devoted to cosmetic, pigmentary, or laser care and post-procedural management.

Participants (Eligibility and Characteristics)

Round 1 (National Survey, $n=601$)

Inclusion: board-certified dermatologists (or national equivalent) in independent practice for ≥ 5 years; $\geq 30\%$ clinical workload in any of the following: pigmentary disorders, cosmetic/laser practice, post-procedure care, or photoaging; routine use or recommendation of dermocosmetics. **Exclusion:** dermatology trainees not yet in independent practice; non-dermatology specialties; questionnaires with $> 10\%$ missing items; duplicate entries (the most complete retained). Captured variables included demographics (age, sex), years in practice, practice setting (academic/public, private, or mixed), region within Egypt, average monthly proportion of clinical caseload for pigmentary disorders and cosmetic/post-procedure care, and frequency of dermocosmetic recommendations.

Round 2 (Expert Panel, $n=16$)

Inclusion: ≥ 5 years in independent practice; recognized expertise in pigmentary/cosmetic/laser dermatology; $\geq 30\%$ clinical workload in the target domains. **Exclusion:** direct commercial employment by a single manufacturer; unresolved conflicts of interest. All 16 panelists were university-affiliated dermatology professors who also maintained active private practice clinics (academic+private). This dual academic-private composition was intentional to reflect both evidence-based and real-world practice patterns. Detailed participant characteristics for both rounds are provided in Supplementary Table S0.

Consensus Procedure

- Round 1—National anonymous ratings ($n=601$): Practicing dermatologists independently rated each ingredient $\times$ scenario on a 1–9 scale (1–3 = inappropriate; 4–6 = uncertain; 7–9 = appropriate). Responses were collected online and fully anonymized. No panel interaction occurred in this round.
- Inter-round feedback (aggregate only): Participants in round 2 received summary, non-identifiable outputs from round 1 (per-item medians and dispersion with scenario histograms). No individual ratings were disclosed.
- Round 2—Expert panel with moderated discussion ($n=16$), then anonymous re-rating: A multicenter senior panel convened for a structured, time-limited discussion focused on discordant or ambiguous items identified from round 1. Moderators followed a prespecified script to minimize dominance, anchoring, and groupthink; no individual round 1 scores were revealed. Immediately after discussion, panelists re-rated all items anonymously using the same 1–9 scale.
- Classification and disagreement rules (a priori): For each ingredient $\times$ scenario, we applied RAND/UCLA rules: median bands 7–9 = appropriate, 4–6 = uncertain, 1–3 = inappropriate; dispersion used inclusive percentiles P30 and P70 to compute $IPR = P70 - P30$; asymmetry $AI = |5 - \text{median}|$; adjusted threshold $IPRAS = 2.35 + 1.5 \times AI$; and disagreement index $DI = IPR/IPRAS$. If $DI > 1.0$, the item was classified as uncertain (disagreement) regardless of the median band; otherwise, the median band determined the class. Final consensus classes were determined from round 2 ratings.
- Materials and independence safeguards: To preserve independence of judgment, no external evidence packets (e.g., literature summaries or guidelines) were provided before rating. Panelists received only the instrument and standardized scenarios. Individual identities were concealed in all analytic outputs.

A schematic of the RAND/UCLA appropriateness workflow (round 1 anonymous $\rightarrow$ inter-round feedback $\rightarrow$ round 2 moderated discussion $\rightarrow$ anonymous re-rating $\rightarrow$ final classifications) is shown in Fig. 1.

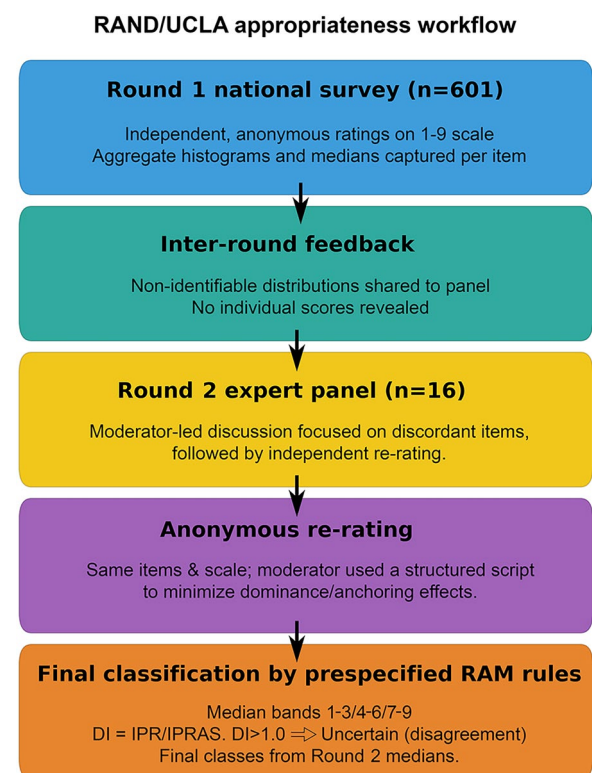


Fig. 1 RAND/UCLA appropriateness workflow: round 1 national survey ($n=601$) → inter-round feedback → round 2 expert panel ($n=16$) with moderated discussion → anonymous re-rating → final classification by prespecified RAM (RAND/UCLA appropriateness method) rules. *DI* disagreement index, *IPR* interpercentile range, *IPRAS* interpercentile range adjusted for symmetry

Outcomes, Data Handling, and Reporting

The primary endpoint was the RAND/UCLA appropriateness class (appropriate/uncertain/inappropriate; uncertain [disagreement] when $DI > 1.0$) per ingredient × scenario in round 2. Final consensus classes were determined from round 2 medians using the rules above. Round 1 distributions (601 raters per item) are reported for transparency and sensitivity and do not alter classifications. Complete round 1 item-level metrics (median, P30, P70, IPR, AI, IPRAS, DI, class) are provided in Supplementary Table S1; round 2 metrics appear in Supplementary Table S2.

Statistics

Analyses were conducted in Python v3.11 using pandas v2.2, NumPy v1.26, and Matplotlib v3.8. Classification strictly followed the RAND/UCLA rules detailed above (medians with inclusive P30/P70, IPR, AI, IPRAS, DI; $DI > 1.0$ labeled uncertain [disagreement]). Final consensus classes used round 2 medians; full round 1 distributions are provided for transparency.

RESULTS

Overview of Agreement and Disagreement

Across ten vignettes and 30 ingredients (300 total items), 158 (52.7%) were appropriate, 135 (45.0%) uncertain—including 47 (15.7%) classified as “uncertain [disagreement]”—and 7 (2.3%) inappropriate. In round 1, 38.3% of items had disagreement ($DI > 1.0$), which fell to 15.7% after round 2 (47/300). Per-item medians and DI values for round 1 and round 2 are listed in Supplementary Tables S1 and S2. The distribution of appropriateness classes across all ingredients–scenario cells is summarized in Fig. 2. Scenario-level distributions of appropriate/uncertain/inappropriate for round 1 and round 2 are shown in Fig. 3.

Handling of High Medians with Disagreement

Several items exhibited a median in the appropriate band yet were classified as uncertain (disagreement) in round 1 because the disagreement index ($DI = IPR/IPRAS$) exceeded 1.0, which overrides the median band under RAND/UCLA rules [8]. This pattern arose where vignette interpretations plausibly diverged across clinical subcontexts. For example, in case 3 (post-procedure skincare), broad-spectrum sunscreen received a high median in round 1 but polarized ratings (immediate post-procedure on a compromised barrier vs. healed phase), yielding $DI > 1.0$ and an uncertain (disagreement) label in round 1. After we standardized the wording in round 2

(initiate photoprotection once re-epithelialized; prefer mineral/tinted as tolerated), dispersion narrowed ($DI \leq 1.0$) and the final class was appropriate. Similar round 1 polarization was seen for classical retinoids (pregnancy/lactation and immediate post-procedure vs. later maintenance use), exfoliants in sensitive/postpartum skin (benefit for dyspigmentation vs. irritation/PIH risk), and selected botanicals (perceived adjunct value vs. concerns about contact irritation/limited evidence). In each case, round 2 clarifications of timing, patient subgroup, or formulation preference reduced disagreement and stabilized the final consensus class (see Figs. 2–3 and Supplementary Tables S1 and S2).

Case 1: Post-Pregnancy Skin Recovery

Clinical Scenario

A 30-year-old woman with Fitzpatrick skin type IV presents with pigmentation on her upper lip, both jawlines, and the lateral areas of her forehead. This pigmentation began during her pregnancy and persisted postpartum, progressively becoming darker over time. The patient is emotionally distressed by the visible pigmentation, which has contributed to her ongoing postpartum depression. For several months, she has tried over-the-counter products without success and acknowledges her lack of adherence to sun protection measures. Considering her status as a breastfeeding mother, which treatment plan will focus on gentle skincare?

Interpretation

Photoprotection (including tinted options) and pigment modulators (e.g., tranexamic acid, arbutin, niacinamide, vitamin C, glabridin) were rated appropriate; classical retinoids and strong exfoliants were inappropriate during breastfeeding because of safety/irritation concerns.

Case 2: Stress-Induced Aging

Clinical Scenario

A 40-year-old man with Fitzpatrick skin type IV presented with concerns about significant skin changes that he attributed to years of high-stress work and poor sleep. His primary complaints included a loss of skin elasticity, deep lines around the eyes and forehead, and persistent dark circles. During the clinical examination, moderate skin laxity, pronounced periorbital wrinkles, and dull, dehydrated skin with uneven texture were observed. He reported minimal use of skincare products but expressed a willingness to comply with a new regimen because he is tired of the comments about his appearance. He mentioned, however, that he is not open to any aesthetic interventions.

Interpretation

Hydration/barrier-repair agents (e.g., hyaluronic acid, peptides, ceramides) were prioritized as appropriate; higher-irritancy actives were uncertain/inappropriate in dehydrated, stress-compromised skin.

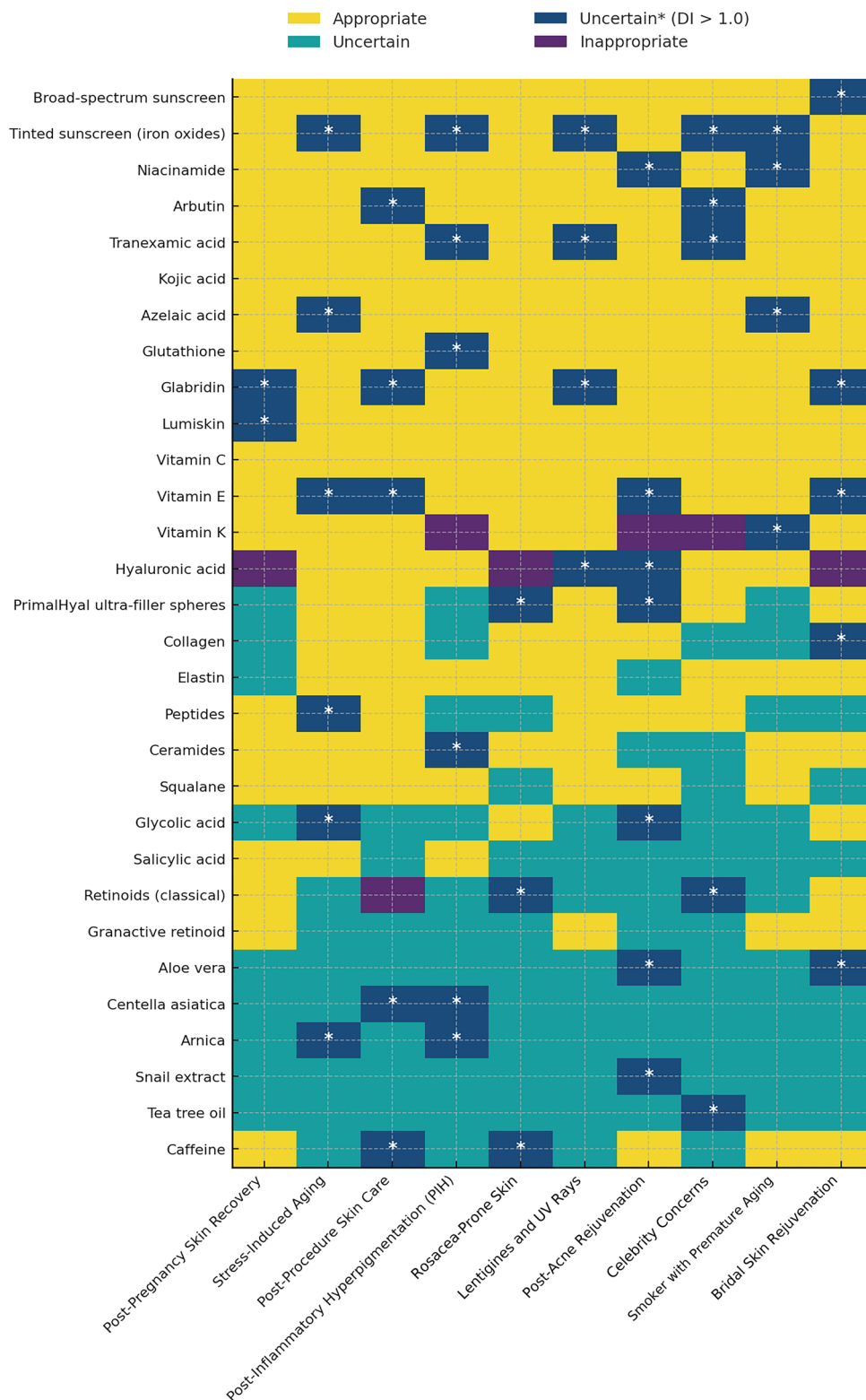
Case 3: Post-Procedure Skincare

Clinical Scenario

A 45-year-old woman with Fitzpatrick skin type III expressed concerns about skin sagging and dryness, which she attributed to hormonal changes related to perimenopause. Upon examination, she exhibited signs of moderate skin laxity, dehydration, and visible fine lines around her mouth and chin. After receiving filler and botulinum toxin injections, she sought advice on post-procedure skincare to maintain her results. She was particularly concerned about keeping her skin hydrated and preventing further signs of aging. What post-procedure skincare would effectively maintain her results?

Heatmap of consensus classes (Round 2)

Asterisk marks items with DI > 1.0



◀**Fig. 2** Heatmap of appropriateness classes for 30 ingredients across 10 scenarios (round 2). Colors encode the three RAM (RAND/UCLA appropriateness method) classes (appropriate/uncertain/inappropriate). Items with disagreement index (DI) > 1.0 are flagged as uncertain (disagreement) and marked with an asterisk

Interpretation

Barrier support and hydration were appropriate; irritants/exfoliants and classical retinoids were inappropriate in the immediate post-procedure window.

Case 4: Post-Inflammatory Hyperpigmentation (PIH)

Clinical Scenario

A 25-year-old woman with Fitzpatrick skin type III presented with concerns about the first signs of aging, particularly fine lines around her eyes and persistent post-inflammatory hyperpigmentation (PIH) from previous acne. She acknowledged having minimal skincare habits and often neglects to remove her makeup before bedtime. This inconsistent routine has likely contributed to her uneven skin tone. The patient expressed interest in preventive strategies for aging and mentioned her curiosity about the “baby botox” trend. However, she is apprehensive because of her fear of needles and concerns about introducing toxins into her body. Upon examination, early signs of collagen degradation were observed around her eyes, as well as pigmented spots on her cheeks and jawline. What are your recommendations for this case?

Interpretation

Brightening/pigment modulators were consistently appropriate; several botanicals were uncertain/inappropriate, often with DI-flagged dispersion.

Case 5: Rosacea-Prone Skin

Clinical Scenario

A 60-year-old woman with Fitzpatrick skin type III, who is prone to rosacea, sought treatment for her increasing wrinkles and facial redness. Upon examination, she displayed redness, visible capillaries, and moderate wrinkles, with particularly delicate skin around her eyes and nose. Her goal for her skincare routine is to address her aging concerns without exacerbating her rosacea or causing further sensitivity. What recommendations would be suitable for her situation?

Interpretation

Non-irritating hydration/barrier repair and gentle antioxidants were appropriate; many exfoliants and potentially stinging actives were uncertain/inappropriate given sensitivity/erythema risk.

Case 6: Lentigines and UV Rays

Clinical Scenario

A 30-year-old woman with Fitzpatrick skin type III and sensitive skin presented with concerns about facial lentigines, fine glabellar lines, and persistent dryness. As a landscape architect, her work involves extensive outdoor exposure, often without adequate sun protection. Additionally, her demanding schedule, balancing work responsibilities and caring for her children, has resulted in her getting less than 5 h of sleep each night. Upon examination, solar-induced lentigines were observed on her cheeks and forehead, along with subtle but emerging fine lines between her brows. Her skin appeared dehydrated, with a rough texture indicating a disruption in the moisture barrier. Chronic UV exposure, sleep deprivation, and environmental stress were identified as contributing factors to her skin concerns. The patient expressed a strong interest in initiating non-irritating antiaging solutions that are suitable for her sensitive

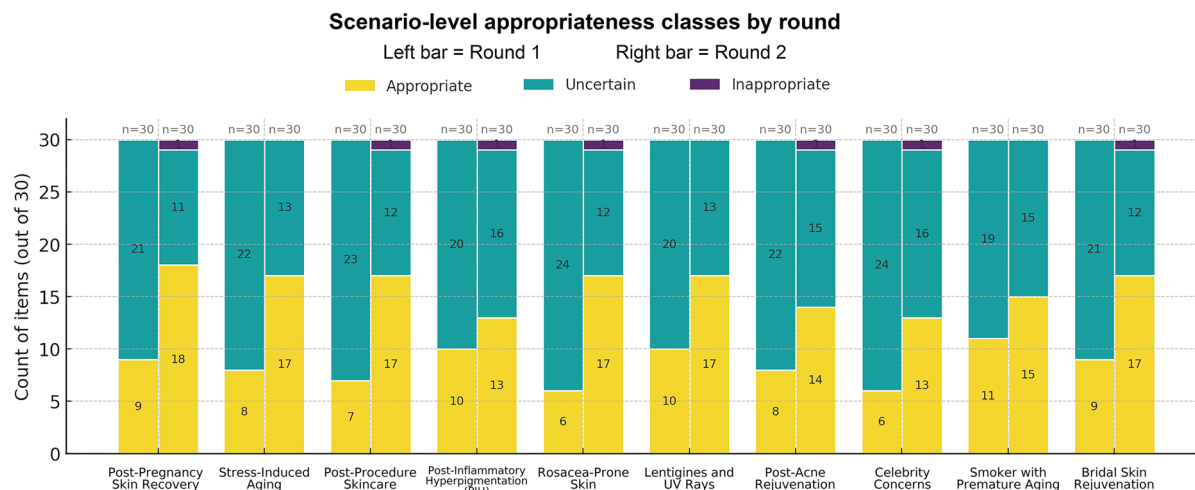


Fig. 3 Scenario-level appropriateness classes by round. For each clinical scenario (30 items = 30 ingredients), grouped bars display the count of items classified as appropriate, uncertain, or inappropriate in round 1 and round 2.

skin, as well as seeking advice on managing dryness and protecting her skin from further sun damage.

Interpretation

Photoprotection was foundational and appropriate; adjunct brighteners and hydration were favored. Irritating actives were scenario-dependent given outdoor exposure and barrier status.

Case 7: Post-Acne Rejuvenation

Clinical Scenario

A 35-year-old man with Fitzpatrick skin type III sought to improve his appearance following recent lifestyle changes. He expressed a desire to resume social activities, which he had been avoiding for the past 6 months because of photosensitivity resulting from a course of orally administered isotretinoin he completed 3 months earlier. Although his cystic acne had significantly improved, he still experienced occasional isolated papules and pustules, along with persistent dryness, particularly around his cheeks and chin. Upon clinical examination, there were signs of mild residual erythema, dry

patches on the face, and subtle post-inflammatory pigmentation from previous lesions. Fine lines were beginning to develop around his eyes, likely worsened by prolonged sun exposure during outdoor activities. The patient wanted to maintain clear skin, address dryness, and implement strategies to prevent photoaging. What dermatocosmetics would align with his specific needs?

Interpretation

Hydration/barrier repair and pigment correction were appropriate; harsher anti-acne botanicals/exfoliants trended uncertain/inappropriate in recently sensitive skin.

Case 8: Celebrity Concerns (Time-Sensitive Public-Facing)

Clinical Scenario

A 40-year-old woman with Fitzpatrick skin type III, living in Cairo, and working as a TV presenter, sought advice on enhancing her skin-care regimen. As a public figure who regularly appears on screen, she felt increasing pressure to maintain a flawless complexion. She had

previously undergone botulinum toxin treatments for forehead lines and hyaluronic acid injections to address volume loss in her nasolabial folds, which had yielded satisfactory results. However, she remained concerned about her uneven skin texture and persistent dullness. Her work environment involved prolonged exposure to bright studio lights and heavy makeup application, often leaving her skin dry and congested. Upon examination, mild skin roughness and subtle pigmentation irregularities were noted.

Interpretation

Focus on texture/radiance with well-tolerated hydrators/antioxidants; avoid over-exfoliation under heavy makeup/studio lights (several exfoliants uncertain/inappropriate).

Case 9: Smoker with Premature Aging

Clinical Scenario

A 45-year-old man with Fitzpatrick skin type IV and a 20-year history of smoking is seeking treatment for premature skin aging. He is concerned about deep wrinkles around his mouth and sagging skin on his cheeks. Having quit smoking a year ago, he wishes to rejuvenate his appearance. Examination reveals deep perioral lines and a dull skin tone. Recommendations for his treatment include fractional laser resurfacing and monthly chemical peels to enhance skin texture. Additionally, botulinum toxin is recommended to address the smoker's lines. Considering his condition, what skincare routine would be advisable for him?

Interpretation

Hydration and antiaging support appropriate; higher-irritancy actives used selectively due to barrier and wrinkle pattern. Pigment correction adjuncts as indicated.

Case 10: Bridal Skin Rejuvenation (Pre-Event Preparation)

Clinical Scenario

A 33-year-old woman with Fitzpatrick skin type III is preparing for her wedding and has expressed concerns about uneven skin tone, fine lines, enlarged pores, and occasional acne. Upon examination, mild pigmentation issues and subtle texture irregularities were noted. Recommendations include a pre-wedding skincare regimen featuring chemical peels and microneedling initiated 3 months prior to the event to enhance skin texture and promote a healthy glow. What do you recommend for such a case?

Interpretation

Photoprotection and hydration as baseline; targeted brighteners appropriate. Irritants timed cautiously around procedures and event-driven timelines.

DISCUSSION

This national consensus study provides insight into the relative appropriateness of dermocosmetic active ingredients across ten clinically relevant scenarios. The results emphasize both universal foundations, such as photoprotection, and context-specific agents that should be tailored to patient needs. Our findings align with international evidence, while also reflecting regionally relevant considerations for patients with higher Fitzpatrick skin types.

Photoprotection was consistently endorsed (median around 8 across scenarios; Fig. 2), underscoring its essential role in pigment and photoaging management with rare DI-flagged uncertain exceptions depending on scenario. This is consistent with robust literature emphasizing photoprotection as the cornerstone of melasma and PIH management [9–12]. In higher phototypes, visible light (VL) contributes to melanogenesis; iron oxide-containing tinted sunscreens mitigate VL-induced hyperpigmentation,

supporting their consistent appropriateness across pigmentation scenarios [4, 7, 9].

Hydration and barrier repair agents, including hyaluronic acid, collagen, peptides, ceramides, and elastin, were prioritized in stress-induced aging and post-procedure care, though round 2 ratings remained scenario-dependent with a few inappropriate classifications. The high appropriateness ratings reflect their ability to improve barrier integrity, restore skin turgor, and support extracellular matrix repair. Notably, PrimalHyal ultra-filler spheres received strong consensus in post-procedure cases, reflecting their dermal plumping and hydration-enhancing properties. While direct clinical trial data on this trade formulation remain limited, its inclusion aligns with the evidence for hyaluronic acid-based interventions in accelerating healing and maintaining skin quality after procedures [1, 2, 13, 14].

Pigment-modulating agents, particularly tranexamic acid, arbutin, glabridin, niacinamide, and vitamin C, were strongly endorsed in melasma, PIH, and chronic sun-induced pigmentation scenarios. Tranexamic acid has accumulated strong evidence as a safe and effective melasma therapy, whether administered topically, orally, or via intradermal injection [10, 11]. Arbutin and glabridin function as tyrosinase inhibitors, while vitamin C provides antioxidant and melanogenesis-suppressing activity, and niacinamide reduces melanosome transfer [15–18]. Asian/Indian melasma consensus and evidence-based pigmentary recommendations similarly prioritize these agents within stepwise treatment frameworks [19, 20]. This evidence base supports the panel's consensus, although careful formulation and long-term adherence remain essential for maintaining pigment control [10, 11, 15–20]. Vitamin K showed scenario-specific behavior—appropriate in lentigines and bridal planning but inappropriate in rosacea-prone, post-acne, and celebrity scenarios—supporting selective, context-dependent use [21].

Retinoids showed divergent appropriateness across cases. Classical retinoids were rated inappropriate specifically in pregnancy, lactation, and the immediate post-procedure scenarios, largely because of safety and irritation concerns. However, granactive retinoid, a

newer retinoid ester, received moderate-to-high appropriateness scores in post-procedure and aging scenarios, reflecting its improved tolerability profile. Retinoids are supported by evidence as gold-standard topical agents for photoaging, collagen stimulation, and keratinocyte turnover [1, 2, 14]. Their cautious but selective endorsement by the panel reflects the balance between efficacy and safety in sensitive populations.

Exfoliating agents such as glycolic acid, azelaic acid, kojic acid, and salicylic acid were inconsistently endorsed. While some utilities were acknowledged in acne and PIH, ratings were low in sensitive, postpartum, and post-procedure cases. This reflects the dual potential of exfoliants: although effective for pigment dispersion and acne therapy, they pose risks of irritation and rebound pigmentation in darker phototypes. Azelaic acid's dual anti-acne and anti-melasma actions and favorable tolerability in Fitzpatrick III–V explain its relatively higher appropriateness compared with stronger alpha-hydroxy acids in sensitive contexts [22, 23].

Botanical and natural extracts, including arnica, aloe vera, tea tree oil, and snail extract, consistently received the lowest scores across cases. Arnica was the only botanical occasionally considered for post-procedure bruising, reflecting its anti-inflammatory potential. However, panelists expressed limited confidence in its role in pigmentary or barrier-repair scenarios. Literature on botanicals in dermatology remains limited and heterogeneous, with risks of irritation, contact dermatitis, and unpredictable efficacy [4, 16, 24]. This likely explains the panel's cautious approach, which mirrors international recommendations that prioritize evidence-based cosmeceuticals.

Within botanicals, *Centella asiatica* showed context-dependent support—aligning with its barrier-supportive and anti-inflammatory profile—yet panel ratings remained largely uncertain outside stressed or recently treated skin. Consistent with our overall stance on botanicals, evidence heterogeneity and limited high-quality trials in higher phototypes tempered strong recommendations [4, 24].

The consensus panel highlighted a selective role for novel and supportive agents in

post-procedural skincare, specifically endorsing granactive retinoid and PrimalHyal ultra-filler spheres, while showing limited support for arnica. Granactive retinoid, a newer-generation retinoid ester, offers the benefits of classical retinoids—including epidermal turnover, collagen stimulation, and dermal remodeling—while reducing irritation risk, which is particularly advantageous in post-procedure recovery [2, 14]. Similarly, PrimalHyal ultra-filler spheres, though trade-specific and less represented in indexed literature, act analogously to cross-linked hyaluronic acid formulations, providing enhanced hydration, dermal plumping, and support for extracellular matrix repair—an effect consistent with the well-established role of hyaluronic acid in improving post-procedural healing and skin quality. In contrast, arnica has been traditionally used for its anti-inflammatory and anti-ecchymotic properties; however, evidence for benefit in pigmentation or structural repair is limited, and trials in post-procedure outcomes are scarce. Accordingly, while granactive retinoid and PrimalHyal spheres may represent promising adjuncts for rejuvenation and barrier restoration, arnica is best considered an optional supportive measure for bruising and inflammation rather than a core therapeutic agent [4, 14, 25].

Taken together, this study highlights several key clinical implications. First, sun protection and hydration are non-negotiable foundations across all dermatologic contexts. Second, pigment modulators such as tranexamic acid and arbutin should be reserved for pigment-dominant scenarios, whereas retinoids require selective, case-specific use. Third, while exfoliants and botanicals retain niche roles, their limited evidence base and safety concerns in darker phototypes preclude their widespread recommendation. The inclusion of novel formulations such as PrimalHyal spheres and granactive retinoid in post-procedure and aging scenarios underscores the importance of evolving dermatocosmetic technologies, though larger, controlled trials remain necessary to validate these findings.

Our classifications largely align with prior guidance while adding region-specific nuance for darker phototypes and high-UV contexts. First, photoprotection was foundational—consistent

with expert recommendations for pigmentary disorders and photoaging—with tinted (iron oxide) sunscreens prioritized where visible light aggravates dyschromia [4, 5, 7, 9, 12]. Second, pigment modulators (e.g., tranexamic acid, arbutin, niacinamide, vitamin C) were appropriate in melasma/PIH-dominant scenarios, aligning with contemporary reviews and Asian/Indian melasma consensus frameworks [10, 11, 15–20]. Third, retinoids retained a central role in photoaging but were avoided during lactation and immediately post-procedure, reflecting tolerability and safety considerations that are often under-specified in prior consensus documents on antiaging dermatocosmetics [1–3, 14, 17]. Finally, botanicals were inconsistently endorsed—our panel took a cautious stance given contact-irritancy and heterogeneous evidence, in keeping with narrative reviews that highlight limited, low-quality data for many so-called natural actives [4, 16, 24]. Table 1 summarizes areas of concordance/divergence and highlights region-specific additions (e.g., visible-light mitigation; timing of actives around procedures; cautious use in sensitive/postpartum skin). These cross-walks support broader relevance and provide a template for replication and benchmarking in other geographies.

This study has several limitations that should be considered when interpreting the findings. First, the consensus ratings were derived from expert opinion rather than randomized controlled trials, which may introduce subjectivity. Although the RAND/UCLA appropriateness method is widely validated for structured consensus generation, the results reflect clinical judgment rather than direct patient outcomes. Second, the study focused primarily on dermatocosmetic agents commonly available in the regional market; newer or less accessible formulations may not have been fully represented. Third, while the ten case scenarios were designed to be clinically relevant, they cannot encompass the full diversity of dermatologic presentations, particularly in different ethnic populations or geographic regions. Another important consideration is the limited availability of high-quality evidence for certain novel or trade-specific formulations, such as

Table 1 Benchmarking the Egyptian RAND/UCLA consensus against prior consensus/guidelines

Domain/ingredient group	Representative prior consensus/guideline evidence	Egypt consensus (this study)	Alignment/divergence
Photoprotection (broad-spectrum and tinted SPF)	Prior expert guidance places broad-spectrum photoprotection as the cornerstone of melasma, PIH, and photoaging, with phototype-adapted advice and emphasis on VL protection in darker phototypes; iron oxide-containing tinted sunscreens are recommended where VL aggravates dyschromia [4, 5, 7, 9, 12]	Foundational across almost all scenarios; VL-mitigating tinted formulas preferred when dyschromia is present	Foundational across almost all scenarios; VL-mitigating tinted formulas preferred when dyschromia is present
Pigment modulators (tranexamic acid, arbutin, niacinamide, glabridin, vitamin C)	Systematic reviews and Asian/Indian melasma consensus documents place tranexamic acid, arbutin, niacinamide, vitamin C, azelaic acid, and related agents centrally in stepwise management of melasma and PIH [10, 11, 15–20]	Scenario-dependent; classified as appropriate when pigmentation predominates (melasma/PIH, chronic photodamage)	Aligned on efficacy; highlights formulation choice, tolerability and adherence in darker phototypes
Hydration/barrier repair (HA, peptides, ceramides, elastin, squalane; PrimalHyal spheres)	Reviews of moisturizers and barrier-repair cosmeceuticals identify hyaluronic acid, ceramides, emollients, and peptides as core components for barrier restoration, turgor, and post-procedure recovery [1, 2, 13, 14]	Prioritized in stress-aging and post-procedure care, with few scenario-specific exceptions. Trade name spheres treated as HA class effect	Aligned on “barrier-first” strategy; clarifies scenarios in which intensive hydration is most strongly prioritized
Retinoids (classical) and retinoid esters (granactive retinoid)	Retinoids remain the gold standard for photoaging and dyspigmentation but require careful titration in sensitive skin, pregnancy/lactation, and periprocedural settings [1–3, 14, 17]	Timing-dependent: classical retinoids inappropriate in lactation and immediately post-procedure; gentler esters (granactive retinoid) endorsed selectively where tolerability is critical	Aligned on central antiaging role; adds explicit timing/contraindication nuance for higher phototypes and post-procedure care

Table 1 continued

Domain/ingredient group	Representative prior consensus/guide-line evidence	Egypt consensus (this study)	Alignment/divergence
Exfoliants (glycolic, salicylic; azelaic; kojic)	Chemical peels and exfoliants improve acne and PIH but carry irritation and PIH-rebound risks in skin of color; azelaic acid has a more favorable profile as a dual anti-acne and anti-melasma agent [22, 23]	Mixed classifications: useful in acne/PIH, but often uncertain/inappropriate in sensitive, postpartum, or immediately post-procedure skin	Qualified alignment; stronger caution signal for darker phototypes and compromised barriers
Botanicals (arnica, aloe, tea tree, snail, <i>Centella</i>)	Narrative reviews describe antioxidant/anti-inflammatory and barrier-supportive potential but stress heterogeneous study designs, small samples and non-trivial irritation/contact-dermatitis risk; few botanicals receive strong graded recommendations [4, 16, 24]	Generally low appropriateness; <i>Centella</i> context-dependent (stressed/recently treated skin); arnica considered optional for bruising only	Conservative stance; mirrors guideline caution given limited, low-quality evidence and irritation risk
Vitamin K	Evidence for topical vitamin K is limited to small studies on post-laser bruising/purpura; there are no strong guideline endorsements for pigmentary or antiaging indications [21]	Scenario-specific: appropriate in selected rejuvenation contexts (e.g., lentigenes/bridal planning); inappropriate in rosacea-prone, post-acne, and celebrity scenarios	Selective endorsement consistent with sparse evidence; reinforces that vitamin K is an adjunct, not a core pigmentary/antiaging agent

SPF sun protection factor, *HA* hyaluronic acid, *VL* visible light, *PIH* post-inflammatory hyperpigmentation

PrimalHyal ultra-filler spheres and granactive retinoid.

Finally, the study population and case scenarios primarily reflect patients with Fitzpatrick skin types III–V, which are highly prevalent in the Middle East and North Africa. While this focus enhances relevance for these populations, the findings may not be directly generalizable to lighter skin phototypes, where tolerability profiles and clinical priorities may differ. Future work should validate these recommendations in prospective cohorts with standardized outcomes, and—critically—report RAM dispersion metrics to enable cross-study comparability.

CONCLUSION

Dermocosmetics function as both foundational and adjunctive elements of care. Photoprotection and barrier support were foundational across most scenarios, while pigment modulators (e.g., tranexamic acid, arbutin, vitamin C) were scenario-specific and most useful when dyschromia predominated. Retinoids—particularly granactive retinoid—were endorsed when tolerability mattered, whereas botanicals had limited support. Across ingredients and vignettes, final ratings showed few DI-flagged disagreements, underscoring the value of context when tailoring dermocosmetic choices for patients with Fitzpatrick skin types III–V. Overall, these findings emphasize the importance of individualized, evidence-informed dermocosmetic selection tailored to clinical context, skin phototype, and patient needs. Integrating these agents strategically alongside medical and procedural interventions has the potential to optimize outcomes, improve tolerability, and enhance patient satisfaction.

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Declarations

Conflict of Interest. Mahmoud A. Rageh, Wael M. Seoudy, Mohamed F. Abozeid, Abeer Mohamed Elkholy, Alaa E. A. Moubasher, Amany Mohammad Abdel-Latif, Dalia Abdel Aziz Attallah, Essamelden M. Mohamed, Hanan Abdelrady Assaf, Hanan Sabry, Mohamed Mahmoud Nasr, Mohamed Saad Hegazy, Nahla Hunter, Nehal Zuelfakkar, Nermeen Ibrahim Bedair, Noha Ezzat Mohammed, Sameh F. Abdelkodous, Yasser Mostafa Gohary, and Sara M. Mohy have nothing to disclose.

Ethical Approval. This study was conducted in accordance with the Declaration of Helsinki and approved by the Ethical Committee of the Faculty of Medicine, Misr University for Science and Technology.

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